



Cr₂O₃-sensitized ZnO electrospun nanofibers based ethanol detectors

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ABSTRACT

Certain amount of Cr₂O₃ has been in situ added to ZnO nanofibers during the electrospinning process and subsequent calcination to form Cr₂O₃-sensitized ZnO (C-s-Z) nanofibers. The morphology and structure of the as-prepared nanofibers have been characterized by scanning electron microscopy (SEM), transmission electron microscopy (TEM), X-ray photoelectron spectra (XPS) and X-ray diffraction (XRD). The effect of the Cr₂O₃ component in C-s-Z nanofibers on the gas sensing properties has been evaluated by the responses to ethanol vapor. The results have showed that the C-s-Z nanofibers containing 4.5 wt% Cr₂O₃ exhibit the best sensing properties to ethanol vapor. The response to 1 ppm ethanol vapor is as high as 3.6, and the response and recovery time are about 1 and 5 s, respectively. In addition, the as-prepared sensors exhibit excellent selectivity and stability. These results indicate that the C-s-Z electrospun composite nanofibers can be used in fabricating high performance gas sensors.

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1. Introduction

Zinc oxide (ZnO), as an interesting chemically and thermally stable n-type semiconductor with large exciton binding energy and large bandgap energy, has been extensively used in the production of photocatalysts [1], optoelectronic devices [2], solar cells [3,4] and so on. Recently, many investigations on sensors based on ZnO for the detection of toxic and combustible gases have also been reported [5–7]. Along with the development of nanoscience and nanotechnology, one-dimensional (1D) ZnO nanostructures have been proposed to be promising candidates for efficient gas sensing [8,9]. In recent years, in order to further improve the sensing properties of ZnO based gas detectors, a series of post-treatment methods for activating ZnO have been developed, including ball-milling [10], sputtering [11], deposition [12], and so on. Noble metal catalysts [6] and some oxides [12,13] are widely used to activate ZnO, and improved gas sensing performances are obtained. However, these post-treatment processes are complicated and suffer from the instability of noble metal modified ZnO based sensors.

Electrospinning is a unique technique which offers a relatively facile and versatile method for the large-scale synthesis 1D nanostructures that are exceptionally long in length, uniform in diameter, large in surface area, especially diversified in composition [14]. Our group has successfully fabricated several efficient ethanol detectors based on single electrospun oxides [15,16]. In the present work, promoted by both the distinct properties of 1D nanostructures

and the effects of oxide additives on the sensing performance, we develop an in situ process for sensitizing ZnO by adding oxides during electrospinning and subsequent calcination. Thus, a new type of ethanol detectors based on Cr₂O₃-sensitized ZnO (C-s-Z) nanofibers is obtained. Although Cr₂O₃ (p-type) is reported to have poor gas sensing properties [17], it can enhance the gas sensing performances of ZnO nanofibers obviously. The as-synthesized sensors based on C-s-Z nanofibers exhibit much more excellent ethanol sensing characteristics, including faster response (~1 s) and recovery (~5 s) behaviors, lower detection limit (1 ppm) and higher sensitivity.

2. Experimental details

2.1. Preparation and characterization of C-s-Z composite nanofibers

Zinc nitrate, chromium nitrate and poly(vinyl pyrrolidone) (PVP, Mw = 1,300,000) were supplied by Beijing Chemical Co. (China). Triton X-100 (C₃₄H₆₂O₁₁) was purchased from Aldrich. Ethanol and dimethylformamide (DMF) were bought from Tianjin Tiantai Chemical Co. (China). All the chemicals were analytical grade and used as received without further purification.

Cr₂O₃-sensitized ZnO (C-s-Z) composite nanofibers are fabricated through electrospinning and followed by calcination. In a typical procedure [18], 0.32 g of zinc nitrate and certain amount of chromium nitrate were added into a mixed solvent of *N,N*-dimethylformamide/ethanol containing a little Triton X-100 as surfactant in a glove box under vigorous stirring for 6 h. Subsequently, 0.6 g of PVP was added into the above solution under

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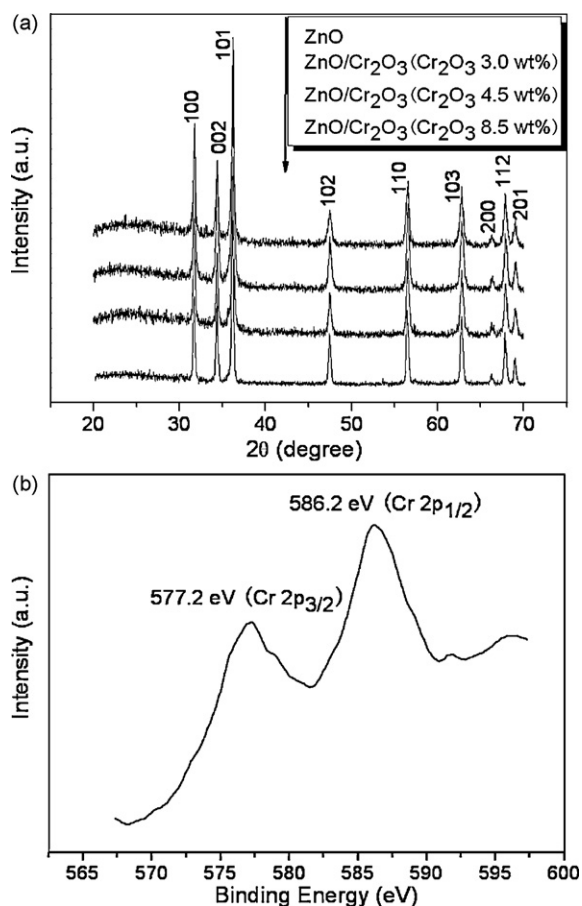


Fig. 1. (a) XRD patterns of the products and (b) XPS spectra of Cr 2p for C-s-Z composite nanofibers containing 4.5 wt% Cr₂O₃.

vigorous stirring for 6 h. Then, the mixture was loaded into a glass syringe for electrospinning by applying a high voltage of 12 kV at an electrode distance of 20 cm. The composite nanofibers were collected on an aluminum frame, transferred to a standard microscopic thin mica slide. After that, the organic constituents were selectively removed from these nanofibers by treating them at 600 °C for 5 h, and crystal C-s-Z composite nanofibers were obtained.

The crystal structures of the products were determined by X-ray powder diffraction (XRD) using an X-ray diffractometer (Siemens D5005, Germany, Cu Kα X-radiation at 40 kV and 50 mA). Analysis of the X-ray photoelectron spectra (XPS) was performed on an ESCLABMKII using Mg as excitation source. The morphologies of the electrospun nanofibers were viewed by scanning electron microscopy (SEM, SHIMADZU SSX-550, Japan). Transmission electron microscopy (TEM, Model JEM-2000EX, JEOL) observations were performed with an accelerating voltage of 200 kV.

2.2. Fabrication and measurement of sensors

The structure of the gas sensor belongs to indirect heating type. The gas sensor is fabricated as follows: the as-prepared sample was mixed and ground with a suitable amount of distilled water in an agate mortar to form a paste. Then the paste was coated on an alumina tube, on which two platinum wires had been previously installed at each end. After the alumina tube was sintered at 300 °C for 2 h, a small Ni–Cr alloy wire was placed through the tube as a heater, which provided operating temperature. In order to improve their stability and repeatability, the gas sensors were aged at 300 °C for 48 h in air.

The sensing properties were tested in a chamber with a volume of 0.001 m³. The target gases were injected into the closed chamber by a microinjector. The response of the gas sensors was measured under a steady-state condition. The electrical properties of the sensors were examined by RQ-2 gas-sensitive capacity measurer. The testing principle of the sensor was similar to that of pervious report [19]. In the present paper, the gas response was defined as $\text{response} = R_a/R_g$, where R_g and R_a denote the sensors' resistances in the presence and in the absence of the detected gas.

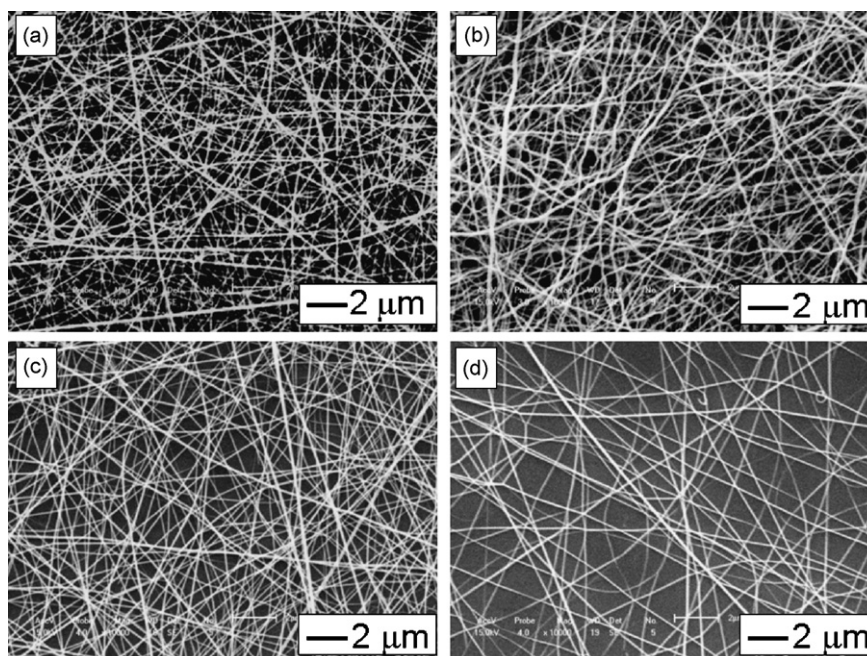


Fig. 2. SEM images of (a) pure ZnO nanofibers and C-s-Z composite nanofibers containing (b) 3.0 wt%, (c) 4.5 wt%, and (d) 8.5 wt% Cr₂O₃.

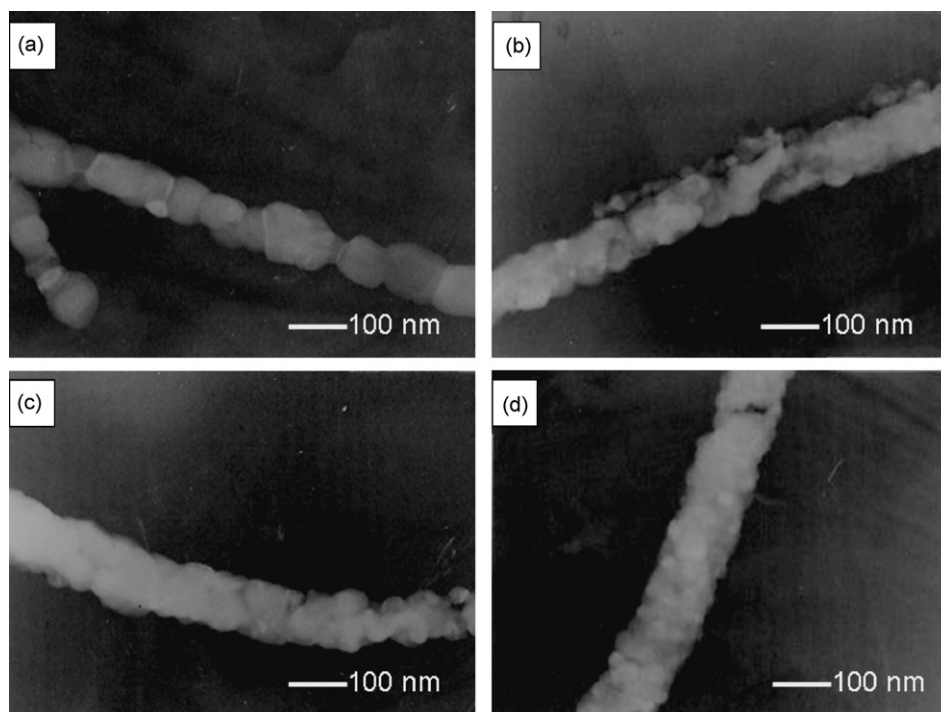


Fig. 3. TEM images of (a) pure ZnO nanofibers and C-s-Z nanofibers containing (b) 3.0 wt%, (c) 4.5 wt%, and (d) 8.5 wt% Cr_2O_3 .

3. Results and discussion

Fig. 1(a) shows XRD patterns of the pure ZnO and C-s-Z nanofibers synthesized by electrospinning technique and subsequent calcination. The observed peaks are matching well with JCPDS reported data of ZnO [20]. The sharp peaks of the XRD patterns correspond to ZnO material and are observed to be polycrystalline in nature. No peaks corresponding to Cr_2O_3 are observed, which may be due to its poor crystallinity and small mass% in the composite nanofibers.

In order to prove the existence of Cr_2O_3 component in the composite nanofibers, further characterization has been done using XPS to investigate the bonding characteristics and oxidation states of Cr in the composite nanofibers. The XPS survey spectrum of the C-s-Z nanofibers containing 4.5 wt% Cr_2O_3 is recorded, which is shown in Fig. 1(b). The Cr 2p spectrum shows two peaks centered at 577.2 and 586.2 eV, respectively, indicating the bonding state of chromium is in the form of Cr_2O_3 [21,22].

The typical SEM images have been used to characterize the morphologies of pure ZnO nanofibers and C-s-Z nanofibers. Fig. 2(a) shows the SEM image of the pure ZnO nanofibers, while Fig. 2(b), (c) and (d) show the morphologies of the C-s-Z nanofibers containing different contents of Cr_2O_3 . From the SEM images, it can be clearly seen that the diameters of the as-prepared nanofibers for all the samples are ranging from 80 to 130 nm. TEM images of the nanofibers are shown in Fig. 3, indicating that the fibers have a rough surface, and formed through the agglomeration of ZnO and Cr_2O_3 nanoparticles.

An effect of Cr_2O_3 component in C-s-Z nanofibers on gas sensing properties has been studied. Fig. 4 depicts the variation of the gas response to 100 ppm ethanol of our nanosensors sensitized by different contents of Cr_2O_3 at 300 °C, which is the optimum operating temperature for our sensors (as shown in Fig. S2, Supporting Information). It is observed that the largest gas response is obtained when the C-s-Z nanofibers containing 4.5 wt% Cr_2O_3 , while the gas response falls down when the content of Cr_2O_3 increases. The response of the C-s-Z nanofibers containing 4.5 wt% Cr_2O_3 is 24

towards 100 ppm ethanol, which is much better than pure ZnO nanofibers. So it is used in all the following experiments.

The sensor based on the C-s-Z nanofibers containing 4.5 wt% Cr_2O_3 exhibits not only extremely high response values, but also ultrafast response/recovery characteristics to ethanol, as shown in Fig. 5. In the measurements, four periods are examined at each fixed concentration. When exposed to 1 ppm ethanol, the response is as high as 3.6. With increasing the concentration of ethanol, the sensitivities greatly increase. The response values are about 5, 13.5, 24, 39 to 10, 50, 100, and 200 ppm ethanol, respectively. These results are even much higher than that of noble metal-functionalized sensors reported previously [6]. Furthermore, it can also be seen that the signal from the sensor becomes stable only within 1 s after it is exposed to ethanol, and returns to the original values within 5 s after the tested vapor is replaced with air. Thus, the response and recovery time are found to be about 1 and 5 s, respectively. We mainly attribute the rapid response and recovery to the 1D nanostructure of our electrospun nanofibers, which can facilitate

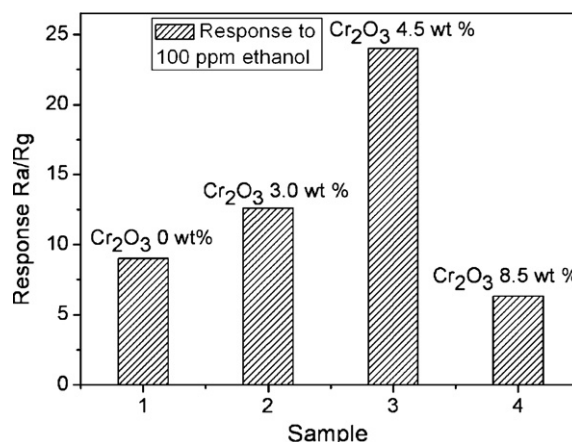


Fig. 4. Variation of gas response with different contents of Cr_2O_3 .

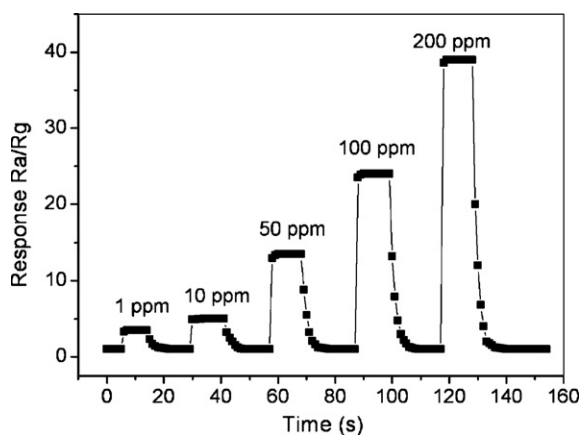


Fig. 5. Response–recovery characteristics of the sensor based on C-s-Z nanofibers containing 4.5 wt% Cr_2O_3 .

fast mass transfer of ethanol molecules to and from the interaction region as well as improve the rate for charge carriers to traverse the barriers induced by molecular recognition along the fibers [18,23].

The responses of the C-s-Z nanofibers containing 4.5 wt% Cr_2O_3 versus ethanol concentrations at 300°C are given in Fig. 6. The response increases remarkably with the increase of ethanol concentration, and when the ethanol concentration is above 15,000 ppm, it reaches saturation. The relationship between the response and ethanol concentration follows the formula $R = AP^\beta$, where P is the ethanol concentration, A and β are constants for certain materials [24]. As shown in the inset of Fig. 6, at relatively low ethanol concentration, the response increases nearly linearly with the ethanol concentration in the range 1–300 ppm, and β is found to be 1. For the pure ZnO nanofibers, however, the sensitivities are only 3.5, 6, 9 and 14.4 to 10, 50, 100 and 200 ppm ethanol. Thus the sensing properties are strongly enhanced by adding certain amount of Cr_2O_3 .

Fig. 7(a) shows the responses of the sensor based on C-s-Z nanofibers containing 4.5 wt% Cr_2O_3 to $\text{C}_2\text{H}_5\text{OH}$ (ethanol), NH_3 , CO , NO , NO_2 , CH_4 and H_2 at 300°C . All kinds of gases are in the same concentration (100 ppm). Those results imply that the sensor is less sensitive to NH_4 , H_2 , CH_4 , and totally insensitive to CO , NO , NO_2 . This trend can be attributed to the effect of operating temperature on the activity of gas molecules [5]. At an operating temperature of 300°C , ZnO-based sensing materials usually exhibit high response to ethanol but very low response to other gases. The long-time sta-

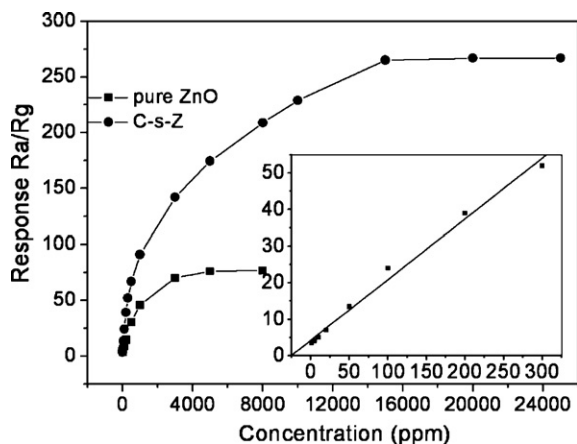


Fig. 6. Response versus ethanol concentration for pure ZnO nanofibers and C-s-Z nanofibers containing 4.5 wt% Cr_2O_3 . The inset shows the calibration curve in the range of 1–300 ppm based on C-s-Z nanofibers.

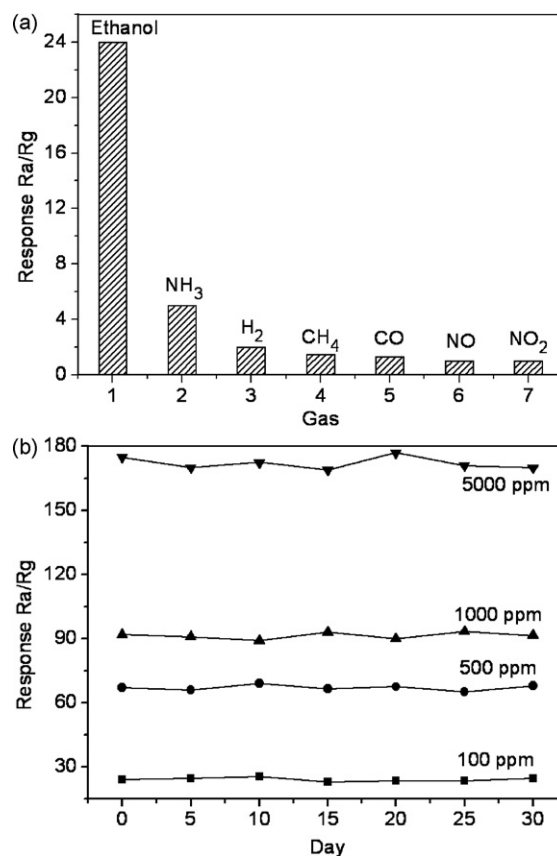


Fig. 7. (a) Selectivity of C-s-Z composite nanofibers containing 4.5 wt% Cr_2O_3 to different kinds of gases, and (b) stability of the sensor after exposing in air for 30 days.

bility of the sensor has also been measured as shown in Fig. 7(b). It can be seen that the sensor exhibits a nearly constant sensor signal to 100, 500, 1000 and 5000 ppm ethanol during the test, confirming the good stability of the sensor based on C-s-Z electrospun nanofibers.

For ZnO-based sensing materials, the change in resistance is primarily caused by the adsorption and desorption of the target gas molecules on the surface of the sensing structure [25]. When exposed to air, the surface of the ZnO nanofibers will adsorb some oxygen molecules. The adsorbed oxygen will capture electrons from the conduction band of the ZnO nanofibers to become oxygen ions. Consequently, depletion layers are formed in the surface area of the ZnO nanofibers with an aligned arrangement, causing the carrier concentration to decrease, and leading to the high resistance of the sensor. When the ZnO nanofibers are exposed to ethanol, the ethanol molecules will react with the adsorbed ions, releasing the trapped electrons back to the conduction band, and then the carrier concentration of ZnO will increase. Accordingly, the resistance of the sensor decreases and the electrical signal can be detected. Comparing with pure ZnO nanofibers, the C-s-Z nanofibers containing 3.0 and 4.5 wt% Cr_2O_3 exhibit much higher gas response, which may be explained as follows. It is observed from Fig. 8 that the initial resistances of C-s-Z nanofibers based sensors are larger than that of pure ZnO nanofibers based in air ambient, which is due to the intergranular potential barrier. It is believed that some heterogeneous intergrain boundaries of Cr_2O_3 (p-type)–ZnO (n-type) can be formed in the C-s-Z composite nanofibers. This leads to the formation of barrier height among the grains and hence increases the nanofibers' resistance in the absence of target gas, which can finally increase the gas response of the as-prepared sensor. Furthermore, Cr_2O_3 nanoparticles dispersed in the ZnO nanofibers may

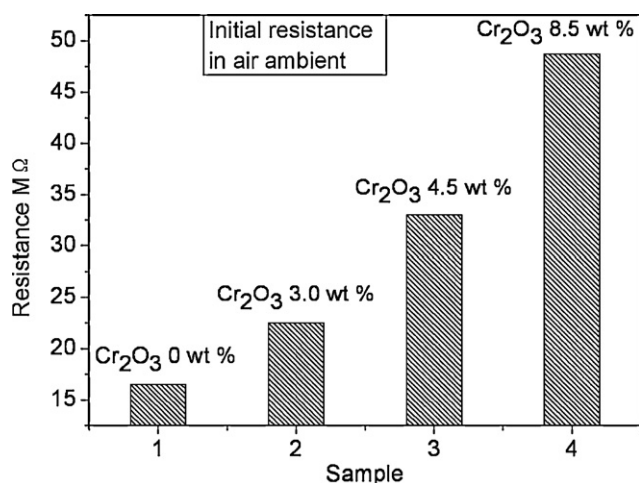


Fig. 8. Initial resistances of the sensors in air ambient.

enhance the absorption of oxygen ions according to the spill-over effect [26,27], resulting in the high gas response of C-s-Z nanofibers. On the other hand, Cr₂O₃ is reported to have poor gas sensing properties, so higher content of Cr₂O₃ would nearly mask the entire base material-ZnO and would resist the gas to reach up to the active sites, then gas response would decrease further [12].

4. Conclusions

In summary, we have developed an in situ process for sensitizing ZnO based gas sensors by Cr₂O₃ additives during electrospinning process and subsequent calcination. With the contribution of heterogeneous intergrain boundaries of Cr₂O₃-ZnO to the sensitivity and spill-over effect, the best ethanol sensing performance, such as higher response, faster response and recovery, lower detection limit, and reproducibility, are obtained for the sensors based on C-s-Z nanofibers containing 4.5 wt% Cr₂O₃. This work presents a simple, effective, and low-cost route to synthesize ethanol sensors. The possibility of large-scale synthesis and excellent sensing properties demonstrate that the C-s-Z nanofibers are very ideal candidates for fabricating ethanol sensors, and the in situ sensitization process by electrospinning will be a promising method for preparing chemical sensors with high performance.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.snb.2009.10.016](https://doi.org/10.1016/j.snb.2009.10.016).

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